Highlights

Playback-Conditioned Joint Prefix-State Repair for Barge-In in Cascaded Spoken-Dialogue Systems

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- A software cursor resolves a legal assistant-token commit boundary.
- KV, mask, token, position, role, and EOT state are repaired jointly.
- All 27 production crops matched direct layer-wise prefix slicing exactly.
- Matched recovery remained exact across 60 steps in the accepted run.
- Fixed-protocol repair avoided long-context re-prefill costs.

Playback-Conditioned Joint Prefix-State Repair for Barge-In in Cascaded Spoken-Dialogue Systems

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Abstract

Barge-in creates asynchronous progress across language-model generation, text-to-speech synthesis, and software playback in cascaded spoken-dialogue systems. We formulate the resulting context correction as external-progress-conditioned joint prefix-state repair. A software-consumed-sample cursor is resolved through text-to-speech fragments to a legal assistant-token commit boundary; the key–value cache, attention mask, token ledger, position progression, content span, role phase, and end-of-turn state are then transformed together. Under a frozen Qwen2-7B-Instruct, Transformers, BF16, and SDPA configuration, an exact-only evaluation covered 24 cases, 27 crop events, and 60 recovery steps. Every production crop was bitwise exact across 28 key–value layers against direct layer-wise slicing of the same pre-crop snapshot, all 27 wrong-length controls were detected, and matched recovery arms remained stepwise exact within the accepted run. A fixed-protocol microbenchmark measured 31.054–48.315 ms for joint crop and role repair over 256–8192-token contexts; the median re-prefill-to-repair ratio increased from 2.254 to 40.620. A downstream fixed-trajectory probe produced generation-minus-playback differences from -3.37 to -1.58 percentage points across four automated operationalizations, with all dialogue-cluster 95% confidence intervals crossing zero. The evidence supports direct crop integrity and within-run matched-arm recovery exactness at a software boundary. It does not establish clean-reprefill equivalence, device or acoustic playback boundaries,

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human-semantic effects, or production end-to-end latency.

Keywords: cascaded spoken dialogue, barge-in, playback cursor, prefix-state repair, KV cache, dialogue history

1. Introduction

Barge-in exposes a state-consistency problem that arises in interactive speech systems. When a user begins speaking while an assistant response is playing, language-model generation, text-to-speech (TTS) synthesis, and software playback may have advanced to different positions. The language model may already have generated a suffix that the synthesizer has only partly converted and the player has not consumed. Committing the complete generated response can make the next turn depend on information outside the observed playback path. Discarding the whole assistant turn instead loses both the retained prefix and its computation. Stopping audio is therefore only one part of barge-in handling: a cascaded system must also decide which assistant prefix remains committed and make every representation of the conversation agree with that decision.

This problem requires a precise account of progress. We distinguish *software-consumed samples*, *device-presented samples*, and *acoustically heard content*. The first quantity is reported by playback software. The second additionally depends on application, operating-system, driver, and device buffers, while the third is further affected by transduction, propagation, and the listener's environment. This work observes only a software-consumed-sample cursor. Accordingly, "playback-conditioned" denotes state operations driven by this software cursor, not token-level knowledge of what reached the device or what a listener heard.

Cascaded automatic speech recognition (ASR)–large language model (LLM)–TTS systems remain attractive because their components can be selected, instrumented, and optimized separately. Recent work has addressed semantic triggering, incremental reasoning, interruption control, and streaming synthesis (Zou et al., 2026; Mai, 2026; Chen et al., 2025; Du et al., 2024). Predictive ASR can also use partial recognition to prepare downstream results before the input is final (Schwarz et al., 2023). Within *Speech Communication*, adjacent studies have examined low-latency integration for telephone-conversation processing (Morrone et al., 2024), the effect of telecommunications latency on speaker transitions (Edwards, 2025), and conversation-history prompting for speech-LLM ASR (Zheng et al., 2026). Faster or earlier computation, however, does not determine which generated content should survive a barge-in.

Playback-conditioned history truncation is established engineering practice. The OpenAI Realtime API accepts an `audio_end_ms` boundary and removes unplayed audio together with the associated transcript (OpenAI, n.d.). Azure Voice Live provides `auto_truncate`, which updates the previous response and session context following an interruption and documents a real-time-playback assumption in its estimate (Microsoft, 2026). LiveKit exposes interruption semantics intended to align stored transcript history with spoken output (LiveKit, n.d.). We therefore do not claim the general principle that history should reflect played output. Nor do we treat cache shortening as novel, since Transformers already provides a `DynamicCache.crop` operation (Hugging Face, 2025b,a).

The research object is narrower and cross-layer: an *external-progress-conditioned joint prefix-state repair contract*. A software cursor is resolved through TTS-fragment metadata to a legal assistant commit boundary. That boundary is then applied consistently to the transformer key–value (KV) cache, attention mask, complete token ledger, assistant-content span, position progression, and chat role/end-of-turn (EOT) state. These components cannot be repaired independently. A cache of the intended length may still encode an invalid conversation if its mask, ledger, positions, or role boundary refers to a different prefix. Conversely, a transcript edit does not ensure that the retained model state represents that transcript.

The contract has four layers. First, *boundary resolution* maps the software cursor through a TTS fragment to an assistant token span. Second, *joint-state representation* treats the cache, mask, ledgers, content span, positions, and role/EOT state as one coupled object. Third, *invariant-preserving transitions* define cropping and role reopening without duplicating or misclassifying structural EOT. Fourth, *falsifiable validation* compares the production crop with direct layer-wise slicing of the same pre-crop snapshot, applies wrong-length negative controls, and checks matched recovery from identical retained states.

The article addresses five questions in decreasing order of evidential importance. RQ1 asks whether joint repair satisfies direct crop integrity and within-run matched-arm recovery exactness under a frozen model/backend, and what its fixed-protocol costs are. RQ2 compares software-cursor retention and complete-generation retention using fixed trajectories and automated information-reproduction probes. RQ3 characterizes how a soft trigger affects candidate availability, discarded-token ratio, and an optimistic synchronous-oracle latency bound. RQ4 audits full-string/full-prefill and segment-wise/incremental implementation paths while recognizing that they are not token-equivalent. RQ5 describes three interrupted-history naturalization implementations, whose present trajectories do not permit a causal comparison.

The sole core mechanism contribution is the implemented path

$$\begin{aligned}\text{software cursor} &\rightarrow \text{TTS fragment} \rightarrow \text{assistant token span}\\ &\rightarrow \text{KV crop} \rightarrow \text{role/EOT recovery.}\end{aligned} \tag{1}$$

Under a frozen Qwen2-7B-Instruct, Transformers, BF16, and scaled dot-product attention (SDPA) configuration (Yang et al., 2024), the accepted exact-only evaluation covered 24 cases, 27 crop events, and 60 recovery steps. The retained production state was bitwise exact across 28 K/V layers against direct layer-wise slicing of the same snapshot, all wrong-length controls were detected, and matched recovery arms remained stepwise exact within the accepted run. These results support direct crop integrity and within-run matched-arm recovery exactness only. Earlier clean-reprefill protocols remain rejected; the accepted protocol does not establish clean-reprefill or continuation equivalence, or correctness across models, backends, or hardware.

The remaining studies provide supporting or exploratory evidence. A fixed-trajectory downstream probe produced uncertain differences across four automated operationalizations and is not a human-semantic or human–computer interaction evaluation. Candidate generation was evaluated in a synchronous segmented-text harness whose timestamps do not establish readiness before a real end of speech or production delivery. A separate implementation-path comparison was not token-equivalent. The paper consequently contributes a testable state contract and bounded software evidence rather than a claim of production barge-in latency or improved user experience.

2. Related work

2.1. Playback-conditioned transcript and history truncation

The closest precedents at the interaction-policy level revise conversation state according to playback progress. OpenAI removes the unplayed audio and transcript suffix at a client-supplied audio boundary (OpenAI, n.d.). Azure similarly updates the response and session context after an interruption (Microsoft, 2026), while LiveKit exposes framework-level semantics for truncating interrupted speech (LiveKit, n.d.). These systems establish that an assistant message need not be retained according to generation completion.

Their public interfaces do not establish their inference internals. The reviewed OpenAI and Azure documentation does not disclose whether the services use a cascaded architecture, how assistant token spans are represented, or whether KV state is cropped in place. LiveKit exposes message and playback behavior but not joint recovery of transformer K/V tensors, masks, token ledgers, positions, and role/EOT state. Our distinction is therefore one of inspectable mechanism and evidence, not an assertion that these systems lack an undisclosed implementation. Their playback quantities are also software-level values or estimates, not direct measurements of device presentation or acoustic reception.

2.2. Streaming dialogue, anticipatory computation, and interruption control

Predictive ASR uses partial recognition to forecast completed input and prefetch a response that can be used if the final recognition confirms the prediction (Schwarz et al., 2023). LTS-VoiceAgent combines semantic triggering with incremental reasoning (Zou et al., 2026), and RelayS2S uses response-level candidate prefixes with verification and continuation (Mai, 2026). These studies establish compute-before-commit as a relevant design pattern for speech interaction. Our supporting candidate study shares that pattern but measures generation before synchronous oracle acceptance. Its candidate-selection event is not equivalent to production availability because cache update, delivery authorization, consumer admission, TTS, and audio buffering may follow it.

Interruption-control research primarily determines when output should stop. FireRedChat combines streaming or personalized voice-activity detection with interruption control in cascaded and semi-cascaded configurations (Chen et al., 2025). In contrast, joint prefix-state repair starts after an interruption decision and determines what part of the assistant turn remains committed. Detection and repair are complementary, but successful detection does not imply a valid next-turn cache and dialogue history.

Full-duplex speech models provide another reference. Moshi models concurrent speech and text streams to support overlapping interaction (Défossez et al., 2024). Tighter stream coordination can avoid some boundaries introduced by independently deployed ASR, LLM, and TTS components, but model-stream position, device presentation, and acoustic reception remain distinct in the presence of queues and buffers. Comparisons must therefore name the observed boundary instead of inferring delivery from an architectural label.

Related studies connect streaming speech processing, conversational timing, and contextual inference. Morrone et al. integrated speech separation and voice-activity detection for low-latency diarization (Morrone et al., 2024); Edwards measured how telecommunications delay changes speaker-transition timing (Edwards, 2025); and Zheng et al. combined conversation history and hotwords for contextual prompting in conversational ASR (Zheng et al., 2026). These works address speech processing, latency, and conversational context, respectively. None of them addresses playback-conditioned transformer-state rollback.

2.3. KV-cache management and cross-turn state

Autoregressive transformers retain key and value tensors for prior tokens to avoid recomputing the full prefix. Transformers exposes this state through cache abstractions, including `DynamicCache.crop` (Hugging Face, 2025b,a). The primitive makes in-place truncation possible but neither selects the conversational boundary nor repairs non-cache state.

Serving research addresses adjacent objectives. PagedAttention organizes KV memory into non-contiguous blocks with demand allocation and copy-on-write (Kwon et al., 2023). SGLang's RadixAttention manages reusable token prefixes in a radix tree (Zheng et al., 2024). These methods improve memory use, throughput, or shared-prefix reuse rather than select an interrupted assistant prefix from software playback progress. IntentKV prunes cross-turn state according to text-agent intent (Li et al., 2026), and Speculative Interaction Agents coordinate provisional results around asynchronous tool calls (Hooper et al., 2026). Their control signals are text intent or tool state, not TTS-fragment ownership and software-consumed samples.

The technical distinction of the present work is consequently not a new cache representation or crop operator. It couples an external software progress signal to a legal conversational prefix, then repairs all representations of that prefix in one transition. Its validation is equally specific: direct slicing of the same pre-crop snapshot checks whether the requested K/V prefix was retained; a wrong-length control checks whether the gate can reject a boundary error; matched recovery checks whether equal retained states remain equal under identical subsequent token chunks and operations. These gates do not establish equivalence to independently recomputed prefixes, semantic correctness of subsequent responses, or portability across inference engines.

A targeted public-source scan was frozen on 3 September 2026. It covered cross-publisher indexes, arXiv, ACL Anthology, ISCA Archive, ACM Digital Library, IEEE Xplore, NeurIPS proceedings, DOI/Crossref metadata, and first-party product and repository sources. Queries combined cascaded speech dialogue, barge-in, anticipatory response generation, cache rollback, and playback-aware history. Within the reported and accessible sources, no publicly inspectable cascaded implementation was identified that disclosed the complete cursor–fragment–token–cache–role path together with reproducible direct-integrity and recovery evidence. Some sources were inaccessible because of automated-access restrictions, and the search was neither a systematic review nor a patent search. This is a bounded non-identification result, not a global-first claim.

3. System model and problem formulation

3.1. Cascaded streaming system

We consider a cascaded spoken-dialogue system

$$\mathcal{S} = \langle \mathrm{ASR}, \mathrm{LLM}, \mathrm{CHK}, \mathrm{TTS}, \mathrm{PLY} \rangle, \tag{2}$$

whose components communicate incrementally. Streaming ASR converts an utterance into stable text segments $U = \langle u_1, u_2, \ldots \rangle$. Downstream processing consumes only final segments rather than mutable partial hypotheses. The LLM incrementally prefills these segments and generates assistant content $Y = \langle y_0, \ldots, y_{G-1} \rangle$, where G is the current content endpoint.

A chunker partitions decoded assistant output into complete text fragments $F = \langle f_1, \ldots, f_m \rangle$ for TTS. Each fragment owns a contiguous, half-open interval on the assistant-content token axis:

$$\begin{gathered} f_j \mapsto [\mathrm{ts}(f_j), \mathrm{te}(f_j)), \\ \mathrm{ts}(f_1) = 0, \qquad \mathrm{ts}(f_{j+1}) = \mathrm{te}(f_j). \end{gathered} \tag{3}$$

A fragment is the atomic unit of retention because the system has no word-, phoneme-, or token-level alignment between synthesized audio and text.

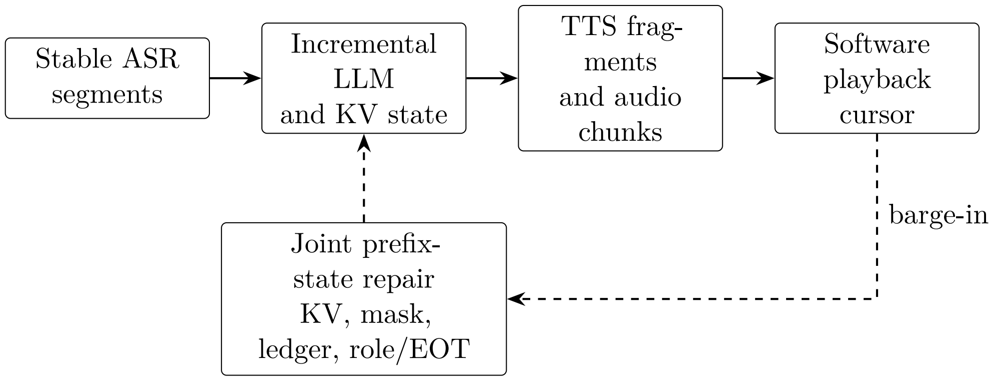

Figure 1: Cascaded architecture and interruption-time state repair. The cursor is a software-level signal; the figure does not imply a device or acoustic boundary.

Streaming TTS transforms each fragment into one or more audio chunks. Once registered, fragment f_j occupies the cumulative sample interval $[\mathrm{ss}(f_j), \mathrm{se}(f_j))$ at sampling rate r. The player reports a cursor $p \in \mathbb{N}$ with counting semantics: $[0, p)$ has been consumed in software.

Figure 1 gives the processing path. ASR segments u_i and TTS fragments f_j are distinct: the former drive input-side prefill, while the latter associate assistant tokens with synthesized samples.

3.2. Software retention boundary

At time t, let $G(t)$ denote the generated-content endpoint and $S(t)$ the largest token endpoint registered on the TTS/software timeline. Because $p(t)$ lies on a sample axis, it cannot be compared directly with $G(t)$ or $S(t)$. If p lies within fragment f_k under the counting convention

$$\mathrm{ss}(f_k) < p \leq \mathrm{se}(f_k), \tag{4}$$

the fragment-level retention boundary is

$$\widehat{H}(p) = \mathrm{te}(f_k). \tag{5}$$

When the cursor falls inside f_k, the boundary is deliberately rounded upward to the complete fragment endpoint. This policy retains the hit fragment's unconsumed tail as well as its consumed prefix; it does not recover an exact transcript of consumed audio. Excluding the hit fragment instead would remove material already consumed in software. Whole-fragment retention

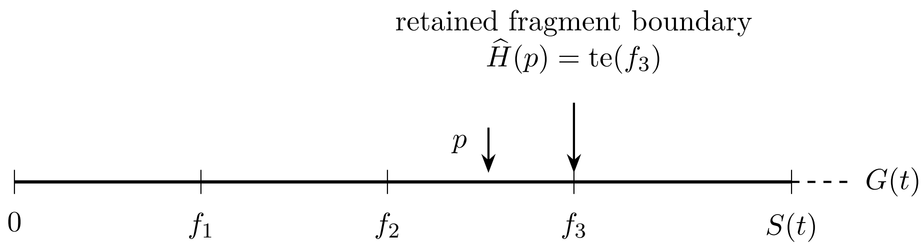

Figure 2: Generated, registered, and software-consumed progress after conversion to the token domain. A cursor inside f_3 selects its complete fragment endpoint.

is therefore an explicit granularity tradeoff, not an empirically established optimal policy. A separate partial flag records the intra-fragment hit without correcting that approximation. For $p = 0$, $\widehat{H}(0) = 0$. A cursor beyond registered audio is clamped to the last fragment with an audio record.

Under the producer-order assumption that playback consumes only registered fragments and fragments enter the timeline in generation order,

$$\widehat{H}(p(t)) \leq S(t) \leq G(t). \tag{6}$$

This is an application-timeline invariant, not a model of acoustic propagation. The reverse query is

$$\Phi : p \mapsto f_k \mapsto [\mathrm{ts}(f_k), \mathrm{te}(f_k)) \mapsto \widehat{H}(p). \tag{7}$$

It is an association across samples, chunks, fragments, and token spans rather than an invertible alignment. Figure 2 illustrates the three progress quantities after they have been mapped to a common token domain.

For downstream analysis only, a partially consumed fragment tail is approximated from the character ratio

$$\alpha(p) = \frac{p - \mathrm{ss}(f_k)}{\mathrm{se}(f_k) - \mathrm{ss}(f_k)}. \tag{8}$$

The raw split $\mathrm{round}(\alpha(p)L_k)$ for fragment length L_k is moved backward to the nearest whitespace boundary. This proxy is not used as a crop point and is neither token-linear nor an acoustic alignment.

3.3. Persistent joint prefix state

Interruption repair is not an operation on a KV object alone. We define the persistent state as

$$\mathcal{Z} = \langle \mathcal{K}, M, I, A, \varphi, e, a_0, a_1 \rangle, \tag{9}$$

where $\mathcal{K}$ is the dynamic KV cache, M is the attention mask, I is the complete token-ID ledger, A is the current assistant-content ledger, φ is the role phase, e is the generation end reason, and $[a_0, a_1)$ is the absolute assistant-content span. Every stable state satisfies

$$|I| = |M| = \mathrm{seq}(\mathcal{K}), \qquad A = I[a_0 : a_1]. \tag{10}$$

Positions for subsequent prefill are derived from the actual cache length after each operation.

The role phase belongs to three states: `USER_OPEN`, `ASSISTANT_OPEN`, and `ASSISTANT_EOT_PENDING`. The end reason explicitly distinguishes none, model-selected EOT, maximum-token termination, consumer stop, and cropping. A selected structural EOT moves the role to `ASSISTANT_EOT_PENDING` and records EOS, but is not forwarded into $\mathcal{K}$, appended to A, or admitted to TTS. A later role-reopening operation is the only close commit and appends exactly one template-derived assistant-close sequence followed by user-open tokens.

For playback interruption, the relative boundary is converted to the absolute endpoint

$$N = a_0 + \widehat{H}(p). \tag{11}$$

The repaired state must shorten K/V, mask, global ledger, assistant ledger, and content span to one semantic boundary; remove all access to discarded content; continue positions from the retained prefix; and assign role/end state consistent with the retained token sequence.

Two zero-content cases are distinct. A playback interruption with $\widehat{H}(p) = 0$ crops to a_0, preserving the assistant header and an open assistant role before normal closure. Full invalidation of an unaccepted candidate instead returns to the pre-speculation `USER_OPEN` endpoint and removes both the assistant header and candidate content. Conflating the two cases either manufactures an empty assistant turn or erases a turn that already entered the output path.

4. Playback-conditioned joint prefix-state repair

4.1. Fragment association and producer contract

Each timeline record contains

$$\langle f_j, [\mathrm{ts}, \mathrm{te}), C_j, [\mathrm{ss}, \mathrm{se}), s_j \rangle, \tag{12}$$

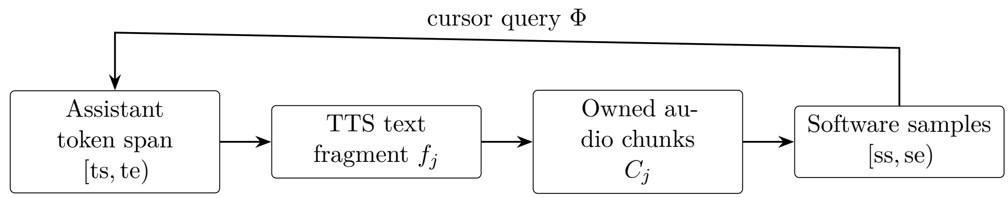

Figure 3: Fragment association record. A TTS fragment links one assistant-token span to its ordered audio chunks and cumulative software-sample interval.

where C_j is the ordered list of audio chunks and s_j is the fragment lifecycle state. Generation registers a frozen token span, TTS attaches chunks and advances the cumulative sample axis, and the player advances p. On interruption, Φ locates the hit fragment and returns its token endpoint (Figure 3).

The reverse query is valid only if write-side constraints hold. Fragment spans must be non-empty, monotonic, and contiguous. A fragment accepts audio only after its span is frozen, and no terminal fragment accepts further chunks. Every chunk identifier is globally unique. Sample intervals must be non-empty, ordered, and contiguous, while the software cursor is monotonic and cannot advance beyond registered samples. Out-of-order fragments, duplicate or misassigned chunks, gaps, overlaps, post-closure appends, and cursor regression are rejected rather than silently reordered.

The chunker sees decoded text rather than token identifiers. To associate a fragment with a token endpoint, the method accumulates non-whitespace characters contributed by each decoded token and locates the fragment's non-whitespace length in that cumulative sequence. Whitespace-only fragments are skipped, endpoints are clamped to the observed content length, and the concatenated non-whitespace fragment sequence must equal the original decoded sequence. The procedure establishes a deterministic software boundary without claiming acoustic alignment.

4.2. Invalidatable candidate generation

Let a turn-completion model assign the cumulative stable transcript a confidence

$$c_i = \mathrm{conf}(u_1 \cdots u_i) \in [0, 1]. \tag{13}$$

When $c_i \geq \theta$, the system stores the current `USER_OPEN` state as a pre-speculation snapshot, opens the assistant role, and generates at most B candidate content tokens. The budget bounds the token volume of a false trigger.

If another stable ASR segment arrives, the active candidate is fully invalidated. The joint state returns to the saved user-open endpoint, including removal of the assistant header, and discarded candidate tokens are counted. Immediately after the crop, `CROPPED` remains observable; successful prefill of new user text resets the current end reason. At the eventual endpoint event, a live candidate can continue or a fresh assistant role can be opened.

The first-candidate callback follows token selection but precedes the cache-update forward and generator yield. It denotes internal selection/compute readiness only. It does not denote delivery authorization, TTS admission, device presentation, or acoustic output. The present harness uses a post-candidate synchronous oracle; an independent online delivery gate is outside the evaluated mechanism.

4.3. Joint crop and role recovery

Given a legal endpoint N, cropping shortens every K/V layer on its sequence dimension and truncates M, I, A, and $[a_0, a_1)$ to the same prefix. It assigns a role phase consistent with the retained structure and marks the current end reason as `CROPPED`. An endpoint inside structural role or EOT tokens is rejected.

For playback interruption, the hit fragment and preceding fragments remain in A, while later content disappears from all cache and ledger representations. When no assistant content is retained, the assistant header remains and the role is still open. The role-reopening operation then appends the template-derived assistant-close and user-open sequence once. If that sequence contains q tokens, the state after reopening satisfies

$$|I| = |M| = \mathrm{seq}(\mathcal{K}) = N + q, \qquad |A| = \widehat{H}(p). \tag{14}$$

The structural tokens enter $\mathcal{K}$, M, and I but not A; the next user text begins at position $N + q$.

Full candidate invalidation instead crops to the saved user-open endpoint before the assistant header and inserts no assistant close. The `CROPPED` state persists until the next user segment is appended. Figure 4 summarizes the playback case and the unique structural close.

Editing text without removing its K/V representation leaves the model conditioned on discarded content. Shortening K/V without truncating the mask and ledgers invalidates lengths and positions. Joint repair makes the same retained prefix inspectable in token space and model state. Here, joint

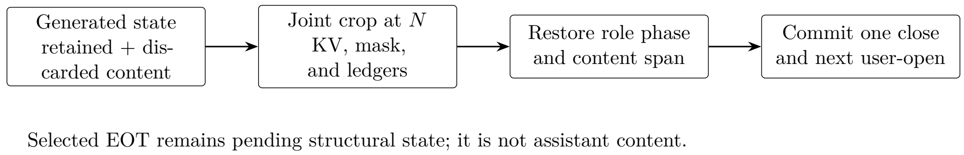

Figure 4: Joint crop and role recovery. A unique close/reopen transition commits structural EOT without duplicating it in assistant content.

consistency is specified at completed operation boundaries; it is not a claim of transactional atomicity for concurrent readers.

4.4. Exact validation

Direct crop integrity is evaluated by comparing three states at every crop event: the prefix extracted from the pre-crop production state, the state after one production crop, and a slicing oracle cloned from the same pre-crop snapshot. The oracle slices every K/V tensor directly to the derived length together with the mask and global token ledger; it does not call the production crop interface. It is nevertheless a same-snapshot oracle, not an external implementation or clean re-prefill reference. It tests preservation of a specified prefix, not whether the pre-crop state was semantically correct or whether a live audio cursor was mapped to the appropriate text. Those boundary assumptions require separate validation.

The frozen grid contains 24 ordered cases covering 512-, 2048-, and 8192-token contexts; zero-content retention; exact and mid-fragment boundaries; reply-tail removal; pending EOT; full invalidation; a later user turn; and a second crop. The cases produce 27 crop events, including three no-ops. All 308 assistant fixture tokens are appended through the production accumulation path.

For each event, the target length is derived from the case, fragment partition, and role/content boundaries rather than trusted from a recorded keep value. Validation compares per-layer K/V shape, dtype, device, SHA-256, and runtime element equality, together with the mask, token ledger, sequence length, content span, role phase, and end reason. A disposable wrong-length control must fail for each event.

After cropping, the production and oracle arms receive identical token-ID chunks and role/state operations for 60 matched recovery steps. K/V, logits, masks, ledgers, retained-prefix hashes, and independently derived role/end/content states are compared after each step. We call this *within-run*

Table 1: Evidence hierarchy and measurement boundary.

Study	Role	Principal limitation
C2 v3	Direct crop and recovery correctness	One snapshot/backend; no clean re-prefill
A1/P1	Fixed-protocol cost and software path	No device or acoustic stopping
E3	Automated downstream probe	No human reference standard
C-E2	Token-consistent candidate characterization	Synchronous oracle
C-E1	Whole implementation-path audit	Paths not token-equivalent
A2	Exploratory history naturalization	Generation trajectories confounded

matched-arm recovery exactness. It means that two exactly matched retained states remain equal under the same operations in one accepted run. It does not mean determinism across processes, devices, versions, precisions, or backends, and it does not establish clean-reprefill or continuation equivalence.

5. Experimental methodology and results

5.1. Evidence hierarchy and environment

The evaluation follows the evidence hierarchy in Table 1. C2 is the core correctness study. A1 and P1 bound model-side repair cost and the prepared-state software control path. E3 probes a downstream automated outcome. C-E2 and C-E1 characterize candidate generation in a separate synchronous harness, while A2 is exploratory. These campaigns have different states and timing boundaries; their absolute times are neither pooled nor subtracted across campaigns.

Data were derived from MultiWOZ 2.1 (Eric et al., 2020). The principal model was Qwen2-7B-Instruct (Yang et al., 2024); TEN Turn Detection supplied turn confidence (TEN Team, 2025); Mistral-7B-Instruct-v0.3 supplied the E3 automated judgments (Jiang et al., 2023; Mistral AI, 2024); a six-sentence CosyVoice2-0.5B profile parameterized deterministic Mock-TTS durations (Du et al., 2024); and Qwen3-0.6B was used by the exploratory rewriter (Yang et al., 2025). The implementation reused Whisper-based ASR (Radford et al., 2023) and stream2sentence version 1.0.0 for output segmentation (Beigel, 2026).

Table 2: C2 v3 exact-gate coverage.

Check	Result
Cases / crop events	24/24; 27/27
Production assistant append	308 tokens
K/V layers compared	28
Recovery steps	60/60
Wrong-length controls detected	27/27
No-op crops	3/3

The frozen runtime used Python 3.10.18, PyTorch 2.8.0+cu128, Transformers 4.57.1, CUDA 12.8, and cuDNN 9.10.2 on RTX 3090 GPUs with 24 GB memory. C2 used BF16 and SDPA. Confirmatory C-E1/C-E2 used two GPUs; other campaigns followed their own manifests.

5.2. C2 exact correctness

C2 v1 and v2 retain their rejected verdicts. The v2 structural checks passed, but its numerical gate passed only 42/45 comparisons and its clean-reprefill control did not match the production forward topology. It could therefore establish neither crop attribution nor clean-reprefill equivalence. C2 v3 posed the narrower, identifiable same-snapshot slicing and matched-recovery questions.

For all 27 crop events, the retained pre-crop K/V prefix, production post-crop K/V, and slicing-oracle K/V agreed across 28 layers in shape, dtype, device, hash, and runtime `torch.equal`. The independently derived keep length, mask, global ledger, sequence length, and K/V length were exact. Across 60 recovery steps, matched arms remained exact in K/V, logits, mask, ledger, retained-prefix hash, and role/end/content state. All wrong-length controls were detected.

These results support direct crop integrity and within-run matched-arm recovery exactness for the tested configuration. They do not revise the rejected v1/v2 verdicts and do not establish clean-reprefill numerical equivalence, continuation equivalence, cross-process or cross-device determinism, or end-to-end production correctness.

5.3. Fixed-protocol repair cost and software path

A1 compared synchronized joint crop-plus-role repair with complete reprefill over 256–8192-token contexts. Each length used five warm-ups and

Table 3: A1 synchronized repair microbenchmark; 50 repetitions per length.

Context length	256	512	1024	2048	4096	8192
Joint median (ms)	31.616	31.852	31.054	31.519	36.903	48.315
Joint IQR (ms)	2.356	2.162	3.099	1.197	0.635	0.928
Re-prefill/joint	2.254×	4.124×	7.707×	15.020×	25.453×	40.620×

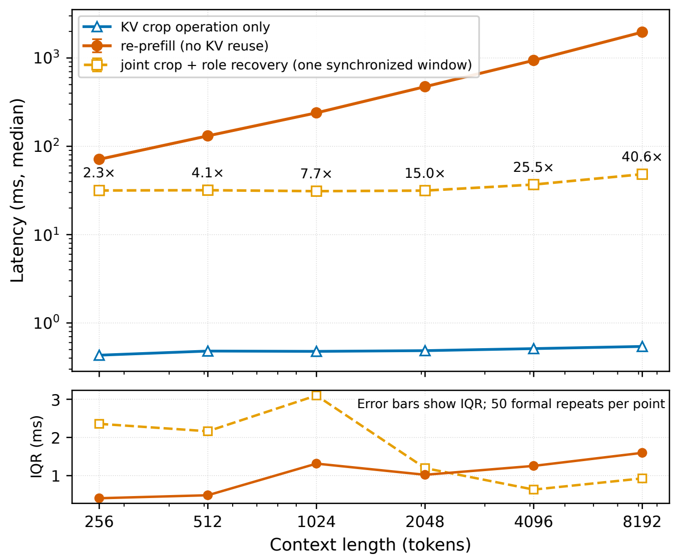

Figure 5: A1 joint repair and re-prefill microbenchmark. Error bars show interquartile ranges across 50 repetitions.

50 recorded repetitions, fixed operation order, and removal of a fixed 32-token suffix. Table 3 and Figure 5 show that joint-repair medians ranged from 31.054 to 48.315 ms. The median re-prefill-to-repair ratio increased from 2.254× at 256 tokens to 40.620× at 8192 tokens. This is a latency microbenchmark, not a state-equivalence test.

P1 crossed three context lengths (512, 2048, and 8192 tokens) with three cursor proportions (0.25, 0.50, and 0.75) and 20 repetitions per cell, for 180 prepared-state headless records. All 180 requests and acknowledgements reached the intended software cursor. A zero leaked-sample count means only that the software counter did not advance after acknowledgement. Cell medians were 0.055–0.062 ms for stop acknowledgement, 0.167–0.176 ms for post-acknowledgement GPU synchronization, 0.47–0.50 ms for timeline lookup, 2.44–2.53 ms from stop to crop completion, and 78.6–80.8 ms from stop to role recovery. The latter intervals are nested. With $n = 20$ per cell, empirical P95 values are descriptive, not service-level objectives. Neither A1 nor P1 includes device/acoustic stopping, online TTS cancellation, or a complete concurrent barge-in path.

5.4. Downstream fixed-trajectory probe

E3 contained 100 dialogues, 400 paired dialogue–injection-position units, 800 condition records, and 1600 judge records. Each assistant trajectory was generated once; software-cursor and generation-retention conditions shared that trajectory, its fragment timeline, and four injection positions (0.25, 0.50, 0.75, and a fragment boundary). Fragment and character-ratio proxy targets were separately eligible when their respective target strings were non-empty. The fragment analysis contained 297 eligible labels from 96 dialogues; the proxy analysis contained 380 labels from all 100 dialogues. Eligibility and uncertainty are therefore target-specific rather than based on 400 independent observations.

The primary estimand assigned equal weight to each eligible dialogue–position label and computed generation minus software-cursor retention. Confidence intervals used 10,000 paired dialogue-cluster bootstrap repetitions. A dialogue-weighted sensitivity first averaged within dialogue. Target-specific exact-key deduplication combined dialogue and trajectory identities, both history keys, and an exact target hash; the fragment key also included the retained token endpoint. It is not semantic clustering.

The lexical detector matched numbers, selected capitalized terms, and non-stopword content terms. The `specific-reference-v3` Mistral judge received the target and two delimiter-joined replies without condition identity, and greedily produced a strict YES/NO decision. Neither detector is a human reference standard. The intervals quantify dialogue-sampling uncertainty conditional on the frozen trajectories, targets, detectors, prompts, and 40-token response cap; they do not include detector error or model/prompt variation.

Figure 6 shows the four label-weighted main-analysis estimates and intervals. Sensitivity results are summarized below. Exact-key deduplication reduced 297 fragment labels to 169 groups and 380 proxy labels to 379 groups. Dialogue weighting yielded effects from -3.21 to -1.30 percentage points; exact-key effects ranged from -2.96 to 0.00 percentage points. All corresponding intervals crossed zero. The four primary estimates therefore es-

Table 4: E3 label-weighted information-reproduction rates. The effect is generation minus software-cursor retention.

Target	Detector	Cursor rate	Generation rate	Effect [95% cluster CI] (pp)
Fragment	Lexical	67.00% (199/297)	63.64% (189/297)	−3.37 [−10.49, 3.40]
Fragment	Judge	42.76% (127/297)	40.74% (121/297)	−2.02 [−10.70, 6.13]
Proxy	Lexical	75.26% (286/380)	73.68% (280/380)	−1.58 [−6.08, 2.67]
Proxy	Judge	43.95% (167/380)	41.32% (157/380)	−2.63 [−8.57, 2.90]

tablish neither advantage nor harm, and they do not establish equivalence, non-inferiority, absence of a difference, or human-semantic effects.

5.5. Candidate-generation characterization

Confirmatory C-E1/C-E2 used 100 unique holdout utterances, five independently initialized process sessions, and ten conditions. Each condition contained 500 session–utterance observations, but the content sampling unit remained 100 utterances. Formal inference used 10,000 crossed/product bootstrap repetitions by independently resampling sessions and utterances and evaluating their Cartesian-product multiplicities.

The harness delivered pre-segmented text synchronously. It excluded real audio, ASR wall-clock time, online turn-detector computation, TTS, playback, and a sound card. `first_token_ready` followed candidate-token selection but preceded cache-update completion and generator yield, so it is reported as candidate selection/readiness. `endpoint_accept` was a post-candidate synchronous oracle, not a natural endpoint detector. The effective-TTFT quantity is therefore an optimistic oracle lower bound.

C-E2 compared the token-consistent B@0.92 and never-trigger paths. Candidate selection/readiness remained between 62.0 and 62.4 ms across all nine B-path points. At B@0.92, candidate availability at oracle acceptance was 67% (335/500), with crossed 95% CI [58%, 76%], and the pooled discarded-token ratio was 2.85% [1.12%, 4.73%]. The latter is a ratio of token counts, not FLOPs, GPU time, energy, or bandwidth. Relative to never trigger, the readiness difference (never minus B@0.92) was −0.03 ms [−0.64,

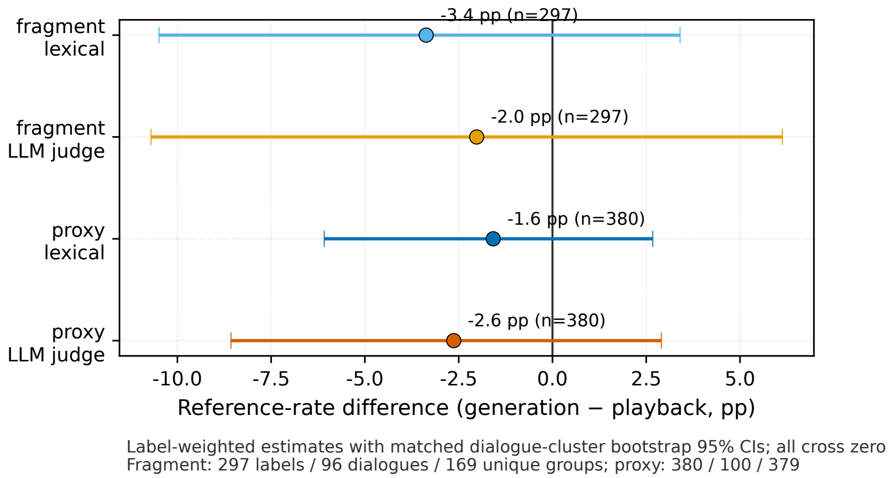

Figure 6: E3 generation-minus-software-cursor effects. Four points and their 95% dialogue-cluster confidence intervals show the label-weighted main analysis only.

0.61], while the oracle-latency lower-bound difference was +20.80 ms [17.85, 23.65]. Across the nine discrete points, lower thresholds increased candidate availability and the pooled discarded-token ratio, while arrival-to-candidate readiness remained between 62.0 and 62.4 ms. These findings characterize generation before synchronous oracle acceptance; they do not show readiness before real end of speech.

C-E1 compared full-string tokenization/full prefill with segment-wise tokenization/incremental forward. The readiness means were 27.70 and 62.38 ms, respectively, giving an A-minus-B difference of -34.69 ms [-35.44, -33.95]. Oracle lower-bound means were 27.70 and 10.26 ms, for +17.44 ms [14.41, 20.32]. Complete token outputs matched in only 280/500 pairs and diverged for at least one session in 44/100 utterances. The contrast therefore combines tokenization, forward topology/shape, role boundaries, kernels, and Python scheduling. It cannot identify an incremental-prefill effect, and selecting only output-matched pairs would be post-treatment selection.

Finally, A2 generated 100 records per naturalization implementation. Mean automated coherence scores were 3.76 for naive retention, 3.62 for lightweight rewriting, and 3.29 for interruption marking. Rewriting took 639 ms on average (median 670 ms; interpolated P90 approximately 935 ms; maximum 1165 ms). Because the conditions regenerated both initial and subsequent responses and shared compatible retained text in only

33/100 dialogues, these values do not estimate a strategy effect. They are exploratory descriptions, not negative or null results.

6. Discussion

6.1. A joint state contract rather than a new crop primitive

The contribution is not playback-conditioned history truncation as a general principle or KV cropping as an isolated operation. Both have precedents. The contribution is an external-progress-conditioned joint prefix-state contract that specifies how an observed software playback position becomes a legal language-model commit boundary and how every prefix-dependent representation is repaired at that boundary.

C2 provides direct evidence for the state-repair part of this contract at the tested fixture-defined boundaries, rather than for the complete speech-to-playback mapping. Under the frozen configuration, every production crop retained exactly the same 28-layer K/V prefix as direct slicing of its own pre-crop snapshot, all known wrong-length perturbations were detected, and matched subsequent operations preserved exact agreement within the accepted run. The evidence goes beyond checking cache length because it covers masks, token ledgers, assistant-content span, role phase, end reason, positions, and post-crop logits. It remains narrower than a clean reconstruction test. The v1/v2 protocols were not reinterpreted after failure; their forward-topology mismatch prevents attribution of their numerical differences to cropping. C2 v3 instead answers an identifiable same-snapshot integrity question.

The most important implication is that transcript correction and cache correction cannot be separated. Text-only truncation can leave discarded content accessible through K/V state, while tensor-only cropping can leave masks, positions, role structure, or audit ledgers at an incompatible endpoint. Structural EOT requires particular care: treating a predicted EOT as assistant content and later adding another close marker silently changes the serialized conversation. Pending structural state and a unique close commit prevent that duplication.

6.2. Supporting results and their limits

A1 shows why retaining a validated prefix can be useful under its fixed protocol. Joint crop and role repair remained in a comparatively narrow time range as context grew, whereas re-prefill increased sufficiently for the median ratio to reach 40.620× at 8192 tokens. The result is not an equivalence proof, and operation order and suffix length were fixed. P1 likewise describes a prepared-state headless path. Its software acknowledgement and crop endpoints do not include TTS cancellation, sound-card buffers, or acoustic stopping. Neither campaign supports a production end-to-end latency claim.

E3 tested whether the selected boundary was followed by detectable information-reproduction differences under fixed automated measurements. All four generation-minus-software-cursor point estimates were negative, but their confidence intervals crossed zero and no substantive margin was prespecified. The result supports neither benefit nor harm and cannot establish equivalence or non-inferiority. Fragment targets depend on TTS segmentation; proxy targets add a character-ratio approximation; the lexical rule and single-model judge operationalize different constructs. Exact-key deduplication removes repeated key weights rather than identifying semantic duplicates. E3 is thus a bounded downstream diagnostic, not evidence of perceived coherence, trust, or dialogue quality.

C-E2 separates candidate computation from acceptance. Its readiness contrast was near zero, whereas its synchronous-oracle lower-bound contrast was positive because an already existing candidate was assigned no post-acceptance wait. Candidate availability of 335/500 uses all condition records as its denominator, not only launched candidates. The discarded-token ratio measures token counts, not compute, energy, or memory traffic. C-E1 is still less isolating because its paths were not token-equivalent. Its timing contrast cannot be attributed to incremental prefill.

The exploratory A2 scores combine different first replies, segmentation boundaries, and subsequent generations. They do not estimate the effect of naive retention, marking, or rewriting. Potential overlap between rewriting and user speech also remains an architectural possibility rather than a measured latency reduction.

6.3. Threats to validity

6.3.1. Construct validity

The software cursor is not a device sample clock or a measurement of acoustic reception. Application queues, audio APIs, operating-system scheduling, drivers, device buffers, and propagation can separate the three boundaries. Consequently, P1's zero leaked software samples cannot be interpreted as zero device-level or acoustic leakage.

The E3 targets and detectors are proxies. Without blinded human annotation or a user study, they do not measure perceived semantic fidelity, naturalness, conversational success, or trust. The synchronous candidate harness likewise measures program events rather than a natural end of speech.

6.3.2. Internal and conclusion validity

C2's exact gate covers a finite deterministic grid and is not an operational error-rate estimate. A1 fixes operation order and crop length. P1 has 20 observations per cell, so empirical P95 depends on one or two upper-tail values. E1/E2 reuse 100 utterances across five process sessions; crossed resampling reflects the two observed dimensions but not day-to-day, load, or deployment variability. C-E1 lacks token equivalence. E3 includes repeated exact keys for some targets, and its intervals do not incorporate detector, prompt, model, or human-judgment uncertainty.

6.3.3. External validity

The evidence is concentrated on English task-oriented MultiWOZ dialogues, Qwen2-7B-Instruct, a ChatML-like role structure, Transformers `DynamicCache`, BF16/SDPA, and RTX 3090 GPUs. Other languages, tokenizers, TTS segmentation, chat templates, precisions, attention backends, and inference engines may define different legal boundaries and state transitions. No experiment includes a real streaming-ASR, asynchronous-TTS, and audio-device loop, loopback waveform, network congestion, or production concurrency. Results therefore apply to the tested software runtime and protocols, not to acoustic stopping, mouth-to-ear latency, production delivery, or user experience.

6.4. Migration conditions

Migration requires more than a cache-cropping primitive. TTS must expose ownership between text fragments and audio chunks; playback must report an interpretable progress position; and the inference backend must retain or reconstruct a legal prefix while jointly updating masks, positions, token ledgers, content spans, role phase, and structural EOT. Each chat template requires its own serialization and close/reopen rules, and each backend requires renewed tests for legal crop points and recovery.

An opaque audio stream requires additional alignment and buffering instrumentation. Word-level durations, device clocks, or loopback waveforms could move the observed boundary closer to device presentation or acoustic reception, but each would change the measured construct. Broader claims would additionally require an asynchronous endpoint gate, TTS admission and cancellation, shared-clock event traces, realistic buffering and concurrency, human annotation, and cross-language or cross-backend replication.

7. Conclusion

This study formulates barge-in context correction in a cascaded spoken-dialogue system as external-progress-conditioned joint prefix-state repair. A software-consumed-sample cursor is resolved through TTS fragments to a legal assistant-token commit boundary, after which the KV cache, attention mask, token ledger, positions, content span, role phase, and EOT state are transformed together.

Under the frozen exact-only evaluation, production cropping was bitwise exact against layer-wise slicing of the same pre-crop snapshot, and matched recovery arms remained stepwise exact after receiving identical token chunks and operations. These findings support direct crop integrity and within-run matched-arm recovery exactness only. They do not overturn the rejected clean-reprefill protocols or establish cross-environment, acoustic, or production equivalence.

The remaining campaigns bound rather than broaden that conclusion. Fixed-protocol measurements show the cost advantage of retaining a prefix over re-prefilling long contexts, while E3 did not determine a stable downstream direction across automated operationalizations. Candidate studies characterize a synchronous oracle and non-equivalent implementation paths rather than benefit before a real end of speech. The evidence therefore shows that external software progress can be converted into a checkable, internally consistent prefix-state transition. Extending that result to device presentation, acoustic reception, other model stacks, production latency, or human interaction requires direct evidence at each corresponding boundary.

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[AUTHOR CONFIRM: declare all relevant financial and non-financial interests for every author, or state: “The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.”]

Data and code availability

The experiment records, manifests, versioned analyses, validation outputs, and available environment snapshots are indexed by the project’s reproducibility record. [AUTHOR CONFIRM: replace this sentence with an immutable public repository or archive URL, release tag or commit, DOI if available, and access date.] The E3 processed input was derived deterministically from MultiWOZ 2.1; the raw MultiWOZ source and model weights are not redistributed and remain subject to their respective licenses. [AUTHOR CONFIRM: verify whether redistribution of the derived E3 input is permitted and add the required attribution or remove it from the public package.] The source repository currently requires confirmation of its rights holder and release license before a public availability claim can be finalized.

Ethics statement

The reported experiments used model inference and text derived from an existing dialogue dataset; the repository contains no direct participant recruitment or newly collected human-subject data. [AUTHOR CONFIRM: verify the target institution’s determination and replace this statement with the required approval, exemption, or not-applicable wording. Confirm separately whether any human annotators participated outside the repository record.]

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